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Integrating Open-Source Process Models into Operational Digital Twins for Grid-Responsive Water Treatment: A Pilot-Scale Comparative Assessment

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ABSTRACT

Desalination and water reuse are energy-intensive, grid-connected processes with untapped potential for demand response. Realizing this potential requires intelligent, self-calibrating digital twins (DTs) capable of closed-loop optimization against dynamic grid pricing. Collaborators at the University of California, Irvine and Oak Ridge National Laboratory developed a closed-loop DT architecture for a reverse osmosis (RO) pilot plant co-located with Orange County Water District's full-scale Groundwater Replenishment System. Their framework integrates cloud-based data acquisition, automated preprocessing, and controller setpoint delivery, employing DYNAMIZU-based process simulation and sparse L1-regularized regression models for operational scheduling. This work integrates DOE's open-source water treatment modeling platform, WaterTAP, as an alternative first-principles modeling core within this established framework. We developed custom workflow layers for physics-based calibration—including online membrane parameter estimation via Pyomo's parmest—and directly embedded the mechanistic flowsheet model (RO with brine recirculation) into multiperiod price-taker optimization against time-varying electricity prices, minimizing total electricity cost subject to cumulative water production targets. Deploying both first-principles and data-driven modeling cores within the same DT architecture and pilot system provides a unique opportunity for direct comparison of these paradigms against common operational data. We present the integration architecture, discuss structural differences in modeling fidelity, extrapolation capability, and optimization integration, and outline a benchmarking methodology with quantitative assessment currently underway. This work demonstrates that the deployed DT architecture can accept open-source first-principles, commercial simulators, and black-box surrogate modeling cores with targeted workflow customization, providing a pathway for broader deployment of grid-responsive DTs across U.S. water infrastructure.

Keywords: Water, Energy Management, Rolling Horizon Optimization, Optimization, Membranes, WaterTAP
additional keywords separated by commas

INTRODUCTION

The water–energy nexus is an important consideration in modern water treatment systems, particularly for energy-intensive processes such as desalination and advanced water reuse. These systems are required to operate under dynamic conditions while maintaining reliable water production. In this context, there is growing interest in adapting system operation to time-varying conditions, such as variation of electricity prices, to improve operational flexibility, enhance system resilience, and reduce overall energy costs.

Digital Twins (DT) enable adaptive and data-driven operation of water treatment systems by integrating real-time data with process models. A key feature is bi-directional communication between the physical system and digital model, allowing continuous system monitoring, model synchronization, and optimization. Recent work at the University of California, Irvine and Oak Ridge National Laboratory (UCI/ORNL) has demonstrated a DT architecture on a reverse osmosis (RO) pilot system within Orange County Water District (OCWD)'s Groundwater Replenishment System by applying real-time adaptive control [1].

Building on these efforts, the framework established by UCI/ORNL integrates data acquisition, preprocessing, calibration, and optimization within a unified workflow to enable adaptive operation. Recent studies have demonstrated its application on a physical pilot system for evaluating DT control strategies, performing high-frequency parameter updates, and enabling autonomous adaptation under dynamic conditions [2–4].

In this work, we extend the UCI/ORNL DT framework by integrating WaterTAP, an open-source modeling platform for water treatment systems. This enables physics-based modeling of the RO process [5,6] within the closed-loop workflow, including online parameter estimation and multiperiod optimization. By deploying WaterTAP alongside existing mechanistic and data-driven models, this work provides an initial comparison across modeling approaches, presenting results in terms of model fidelity and computational performance [7].

METHODOLOGY

Pilot Description

Figure 1 shows a simplified process diagram of the RO pilot installed at OCWD's Groundwater Replenishment System (as modeled in WaterTAP). Processes omitted from the diagram include UV reactors located (1) between the ultrafiltration (UF) module and the cartridge filter (CF) pump and (2) on the RO permeate line; a storage tank for UF permeate, located right before the CF pump, is also not accounted for. The UF pump is a constant-speed pump that is either on or off. Similarly, the CF pump is a constant-speed pump, while the RO pump is driven by a variable frequency drive. RO brine is recirculated to the inlet of the RO pump. The control setpoints are the permeate, recirculated brine, and reject brine flowrates. Power is measured cumulatively for the whole pilot. Additionally, temperature, pressure, flowrate, and conductivity are measured for relevant streams.

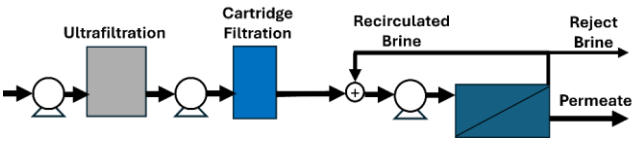


Figure 1. Simplified process diagram of pilot system (physical twin), as modeled in WaterTAP

Model Formulation

The 0D RO model in WaterTAP [7] was used in the process flowsheet. Zero-order models for UF (UF) and cartridge filtration were added. Initially, a naïve assumption of constant lumped efficiency at 40% was assumed for the CF pump and RO pump. The UF pump was assumed to operate at constant power. Subsequently, using PySMO [8], we developed surrogate models to represent variable efficiency of pumps and motors. We achieved this by digitizing the performance curves for both RO and CF pumps and their associated motors, allowing us to model power usage as a function of flowrate and/or head. For the RO pump, we applied pump affinity laws to determine performance curves at alternative speeds, given that the manufacturer's data only covered performance at rated speed. Upon constructing and solving the steady-state, mechanistic flowsheet, we used the PriceTaker class [9] in IDAES [10] to build a multiperiod version of the model. Notably, while RO membrane parameters were calibrated through online parameter estimation, power was not calibrated from measured power data.

Comparatively, a process-based DT was also developed in DYNAMIZU to represent the physical twin from feed pumping through UF, UV treatment, break tank, cartridge filtration, and the advanced RO unit with recirculation. The DYNAMIZU formulation used mass-balance relationships to connect feed, permeate, concentrate, membrane feed, reject, and recirculation flows, while plant power demand was represented through pump power and mixing-energy terms. To align the model with the physical pilot, feed composition and energy-related parameters were calibrated against measured true power using a Python-based optimization routine. Lastly, a data-driven DT was developed in Python using an Empirical Model-Based Optimization (EMBO) framework. In this approach, measured permeate, concentrate, and

recirculation flowrates were used as inputs to an adaptive sparse surrogate model for aggregate pilot power demand. Polynomial features up to cubic order were considered, and L1-regularized regression with cross-validation was used to retain an interpretable set of active terms while limiting overfitting as data accumulated.

DT Framework

Figure 2 shows the DT framework implemented in this work.

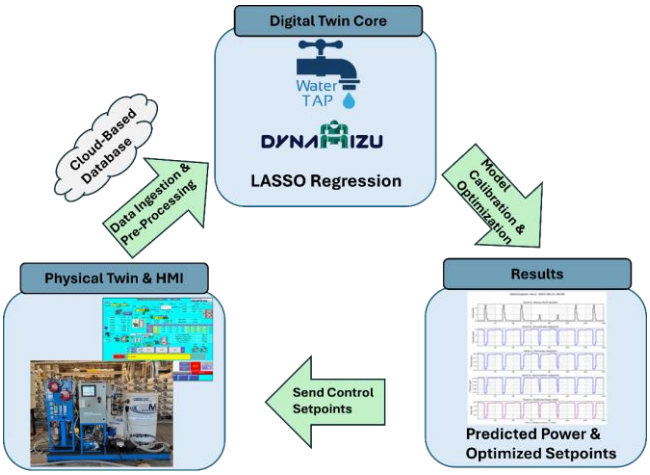


Figure 2. Bi-directional digital-physical twin framework for real-time energy-aware operation of a water reuse pilot.

A common DT workflow was leveraged across WaterTAP, LASSO, and Dynamizu models, for the most part. In the case of WaterTAP-driven experiments, there are some differences to consider. Listed below are the major steps taken in each DT loop when using WaterTAP:

1. **Data ingestion –** all modeling approaches leverage the same or similar code to download CSV data from cloud-stored pilot data. One change made in the WaterTAP approach was to avoid conversion to an XLSX file, which takes more time to process and read. As a general exercise, we attempted to minimize processing time when possible. Rather than aggregate the last 7 days of data to then use for calibration later, we simply pull the latest available CSV file from the cloud.
2. **Preprocessing –** the ingested data is preprocessed via three sequential filters: temporal downsampling at the resolution desired (e.g., raw resolution of 4-second measurements resample to 1-minute average), steady-state detection via coefficient-of-variation filter on key variables (e.g., flowrates, pressure, recovery), and time-weighted sampling to optionally weigh recent data more significantly; exponential decay weighting was applied for this purpose, since these sampled data would primarily be used in the next step for calibrating the RO models.
3. **Calibration –** the preprocessed data is fed into a parameter estimation workflow using Pyomo's parmest [11] to estimate the water and salt permeability coefficients of the RO module while minimizing the sum of squared error for permeate flowrate and concentration. Rather than the full flowsheet, a simplified flowsheet with only the feed composition and RO unit defined was used for speeding up the parameter estimation step. Notably, we elected not to calibrate predicted power, as an exercise to see how well the model could later predict power based on claimed performance from equipment datasheets.
4. **Optimization –** IPOPT [12], an interior point solver, and the default solver used in IDAES and WaterTAP, is used with the MA57 linear sub-solver to minimize energy cost over the time horizon, subject to meeting a target water production over that horizon, while satisfying all physics-based constraints within the flowsheet model. Prior to entering the loop, the full multiperiod model is built once and cloned across the remaining 167 periods (168 timesteps for hourly resolution in 7-day horizon). This model is cached and re-used for all iterations, which avoids expensive model build time. Each iteration updates only the membrane parameters and electricity prices before re-solving.
5. **Send control setpoints and save model data –** The optimal flowrates for the next hour (permeate, recirculate, and concentrate) are extracted from the schedule and written to an Excel file mapped to programmable logic controller (PLC) tag names, from which an MQTT service publishes the setpoints to the pilot plant controller. In shadow mode, this write is bypassed and the setpoints are logged only. Each iteration also saves the full 168-hour schedule, a JSON snapshot of the Pyomo model state for warm-start recovery, and a calibration summary recording water permeability coefficient, salt permeability coefficient, total energy cost, and total water production, all timestamped for post-experiment traceability.

WaterTAP-driven experiments were conducted in February, 2026, over the span of roughly 7 days. However, much of this period was spent on testing and revising the code while debugging issues that arose during live experiments. Thus, in order to run an hourly, 7-day price signal into the model and control the pilot based on these prices, experiments had to be compressed to represent the full 7-day price signal. The longest experiment that was conducted successfully lasted for 14 hours, where every iteration was run at a cadence of 5-minutes (i.e., every 5-minute iteration was representative of an hourly time step in the 7-day horizon), allowing for the full 168 time-step horizon to be simulated within live DT



Figure 3: Measured true power and predicted power consumption via constant efficiency assumption and after accounting for variable efficiency of pumps and motors (top). Corresponding power residuals are shown for constant efficiency ("Old residual") and variable efficiency ("New residual") predictions (second row). Estimated breakdown of power consumption between baseline consumption, UF pump, CF pump, and high pressure (HP) RO pump (third row). Hourly electricity prices that the DT responded to while optimizing energy costs (bottom).

experiments under time limitations.

RESULTS AND DISCUSSION

Closed-loop operation and power prediction

With a constant lumped pump and motor efficiency (40%), the model predicted total pilot power with a mean absolute percentage error (MAPE) of 26% against measured power; predicted power was predominantly underestimated. Replacing the constant-efficiency assumption with digitized pump and motor surrogates, we reran the model retrospectively against the same flowrate setpoints and membrane parameters identified during the online experiment. Consequently, this reduced the MAPE to 6.7% (Figure 3). No online calibration of measured power was performed at any point; the model's power prediction rests entirely on manufacturer datasheets and the membrane parameters identified from permeate flow and concentration. On the other hand, the LASSO model, which does leverage online calibration for power predictions, achieved a MAPE of 1.71%.

Additional information provided by using the first principles WaterTAP flowsheet includes power consumption and efficiency estimates for each pump and motor. However, surrogate representations of pump and motor power can inhibit solver convergence when flowrate setpoints exceed the defined bounds of these surrogates, which are taken directly from equipment datasheets. There is also underlying uncertainty attributed to actual equipment performance in comparison to claimed performance, parasitic power losses (e.g., sensors panels and auxiliary loads) that were assumed constant, and frequent pump shutdowns that were triggered when the UF feed tank approached its minimum or maximum level—all factors that call for the incorporation of power calibration to improve accuracy.



Figure 4: Computational time per cycle during the online, WaterTAP-driven experiment and associated breakdown for data ingestion, preprocessing, calibration, optimization, and model saving/setpoint writeback steps (top). Computational time breakdown per cycle shown as percentage of cycle time (center). Isolated computational time per cycle for the optimization step, which is the most costly step involved (bottom).

Computational performance

Over the 14-hour experimental run using WaterTAP, the median cycle time was approximately 50 seconds. The optimization step comprised 44.4% of the median cycle time (18 seconds, with a range of 13 to 483 seconds), followed by data ingestion at 26.1%, model saving and setpoint writeback at 19.6%, model calibration at 7.9%, and preprocessing at 1% (Figure 4). As per the median computational time per-cycle for WaterTAP, LASSO, and DYNAMIZU models goes, while the LASSO model can be calibrated more quickly per cycle (1.2 s vs 3.2 s), the WaterTAP model outperforms during the optimization step and is 10 times faster than the optimization step when using the LASSO model (18 s vs 183.2 s). Using the mechanistic model via DYNAMIZU could take on the order of hours to optimize, thus the decision to develop a surrogate model of the DYNAMIZU flowsheet, which still results in a lengthier optimization step than both the WaterTAP and LASSO approaches.

In terms of data requirements to achieve usable predictions, the WaterTAP model can immediately yield accurate predictions, as it relied on equipment datasheet performance to estimate power as opposed to calibrating predicted power to measured power. LASSO and DYNAMIZU approaches leveraged the previous 7 days of hourly pilot data to recalibrate power predictions at every hourly interval. Notably, the LASSO approach is able to yield usable predictions starting at the fourth iteration.

Structural differences across modeling cores

In terms of modeling fidelity, WaterTAP's first-principles model achieves 6.2% MAPE without being calibrated to measured power data. On the other hand, the LASSO model is trained on measured power data and achieves 1.71% MAPE; the DYNAMIZU model leverages a surrogate model derived from its first-principles formulation and is also calibrated by measured power data. With regard to extrapolation capabilities, the WaterTAP and DYNAMIZU models are physics-constrained, facilitating extrapolation beyond the boundaries of the training dataset, although the DYNAMIZU approach depends on training data for power calibration; the LASSO approach can interpolate within its available training dataset but can face limitations when attempting to extrapolate beyond the bounds of the training dataset. Finally, optimization is integrated into the WaterTAP approach; the WaterTAP flowsheet is formulated directly as a system of algebraic constraints in a multiperiod price-taker problem, with parameter updates passed through to subsequent solves within the same equation-oriented model; the LASSO and DYNAMIZU cores require external optimization loops wrapped around their predictions.

CONCLUSION

We integrated WaterTAP as a first-principles modeling core into an existing closed-loop DT framework for an RO pilot, alongside the framework's existing commercial-simulator and data-driven cores. The WaterTAP-driven loop was demonstrated over a 168-step horizon in 14 hours of live operation at a median per-iteration cost of ~50 seconds. This uncalibrated physics-based power model—predicted from manufacturer datasheets, with membrane water and salt permeability coefficients identified from RO performance data—achieved 6.2% MAPE against measured pilot power once variable pump and motor efficiencies were modeled. Quantitative and qualitative benchmark comparisons with the commercial-simulator and data-driven cores were presented. Future work should extend the quantitative cross-model assessment through longer-duration live experiments, and conducting out-of-envelope extrapolation test are the immediate next steps.

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DECLARATION OF USE OF AI

AI was not utilized in the preparation of this manuscript.

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